

Lesson 4 Problem 5

The sample mean of the original data is, $\bar{y} = \frac{1+2+3}{3} = 2$

For bootstrap resamples of size 3 from $\{1, 2, 3\}$, the sample mean values and their frequencies among the $3^3 = 27$ ordered resamples are:

1 - 1 time	7/3 - 6 times
4/3 - 3 times	8/3 - 3 times
5/3 - 6 times	3 - 1 time
2 - 7 times	

So the bootstrap distribution of the mean has cumulative probability:

1, $\frac{4}{3}$, $\frac{5}{3}$, 2, $\frac{7}{3}$, $\frac{8}{3}$, 3

with cumulative masses,

$\frac{1}{27}$, $\frac{4}{27}$, $\frac{10}{27}$, $\frac{17}{27}$, $\frac{23}{27}$, $\frac{26}{27}$, 1.

An 80% percentile bootstrap interval keeps the middle 80%, so we cut off 10% in each tail:

10th percentile: $\frac{4}{3}$

90th percentile: $\frac{8}{3}$

$$\Rightarrow \left[\frac{4}{3}, \frac{8}{3} \right] \approx [1.33, 2.67]$$

Lesson 5

- (2) Let the model be $y_i \approx \beta_1 x_i$ with no intercept.

(a) The ridge objective is

$$\min_{\beta_1} \sum_{i=1}^n (y_i - \beta_1 x_i)^2 + \lambda \beta_1^2$$

If the course uses $\frac{1}{n}$ convention, then it would be,

$$\min_{\beta_1} \frac{1}{n} \sum_{i=1}^n (y_i - \beta_1 x_i)^2 + \lambda \beta_1^2,$$

which gives the same minimizer up to how λ is scaled.

(b) Expand and differentiate:

$$\begin{aligned} \sum_{i=1}^n (y_i - \beta_1 x_i)^2 + \lambda \beta_1^2 \\ = \sum_{i=1}^n y_i^2 - 2\beta_1 \sum_{i=1}^n x_i y_i + \beta_1^2 \sum_{i=1}^n x_i^2 + \lambda \beta_1^2 \end{aligned}$$

set derivative to zero:

$$-2 \sum_{i=1}^n x_i y_i + 2\beta_1 \left(\sum_{i=1}^n x_i^2 + \lambda \right) = 0$$

$$\text{So, } \hat{\beta}_1 = \frac{\sum_{i=1}^n x_i y_i}{\sum_{i=1}^n x_i^2 + \lambda}$$

③ Using the usual negative log likelihood plus ridge penalty, the objective is,
(a)
$$J(\beta_0, \beta_1) = - \sum_{i=1}^3 [y_i \log p_i + (1-y_i) \log(1-p_i)] + \lambda \beta_1^2$$

since $\lambda = 1$, this becomes

$$J(\beta_0, \beta_1) = - \sum_{i=1}^3 [y_i \log p_i + (1-y_i) \log(1-p_i)] + \beta_1^2$$

(b) Compared with the unregularized model ($\lambda = 0$):

- Bias increases
- Variance increases.

So the regularized logistic regression has higher bias but lower variance than the unregularized model.

⑤ (a) Valid - It shrinks β_2 and is convex.
Inappropriate: only penalizes one coefficient.

(b) Invalid; odd powers can be negative, not proper shrinkage.

(c) Valid; convex, nonnegative.
Not ideal: treats features unevenly; doesn't address x_1-x_3 dependence.

(d) Valid; convex, nonnegative.
Not ideal: uneven penalties; ignores collinearity.

(e) Invalid; not convex.

(6) Let,

$$\eta_{ik} = \beta_{0k} + \beta_{1k}x_{i1} + \beta_{2k}x_{i2},$$

$$p_{ik} = \frac{e^{\eta_{ik}}}{\sum_{r=1}^3 e^{\eta_{ir}}}, \quad k = 1, 2, 3.$$

Then the λ -penalized negative log-likelihood is;

$$J(\beta) = -\sum_{i=1}^4 \log p_{i, y_i} + \lambda \sum_{k=1}^3 (|\beta_{1k}| + |\beta_{2k}|)$$

For this data, the class labels are $y = (1, 2, 3, 2)$, so explicitly:

$$J(\beta) = -\log p_{11} - \log p_{22} - \log p_{33} - \log p_{42} \\ + \lambda \sum_{k=1}^3 (|\beta_{1k}| + |\beta_{2k}|).$$

⑦ (a) Yes, it changes the least squares solution.

The LS estimate has

$$\beta_1^2 + \beta_2^2 = (6.36)^2 + (8.53)^2 \\ \approx 113.21 > 16,$$

so it lies outside the ridge constraint region. The constrained solution must move inward, so it is shrunk.

(b) No, it does not change the least squares solution.

The LS estimate has

$$|\beta_1| + |\beta_2| = 6.36 + 8.53 \\ = 14.89 \leq 25,$$

so it already satisfies the Lasso constraint. Therefore the solution stays the same.

① Using $TSS = (n-1)s^2$

$$= 14 \cdot 1000 \\ = 14000$$

we get $RSS_d = 14000 - RegSS.$

(a) Mallows' C_p : select $d = 1$
AIC : select $d = 1$
Adjusted R^2 : select $d = 2$

(b) For the $d = 2$ model, the overall F-test is,

$$F = \frac{\text{RegSS} / (d-1)}{\text{RSS} / (n-d)} = \frac{1011 / 1}{12989 / 13} \approx 1.01.$$

With $df_1 = 1$, $df_2 = 13$; this is far below the 0.01 critical value, so we fail to reject H_0 .

So the selected model is not overall significant at $\alpha = 0.01$.

Lesson 6 Problem 1

- (a) For the 6 labels, there are 3 ones and 3 zeros, so the root entropy is,

$$H = -\frac{3}{6} \log_2 \frac{3}{6} - \frac{3}{6} \log_2 \frac{3}{6} = 1 \text{ (bit)}$$

If you use natural logs, it is $\ln 2$.

- (b) Best first split: $X_1 \geq 6.5$

This gives one pure node (1,1) and one mixed node.

- (c) Second split:

for the node with $X_1 < 6.5$, split by $X_2 \geq 1.5$.

This isolates the single positive example.

For the node with $X_1 \geq 6.5$, no split is needed since it is already pure.

- (d) Yes. A non trivial tree can still have zero information gain if each split leaves the class proportions unchanged in the children, so the weighted child entropy equals the parent entropy. The tree splits, but it does not reduce uncertainty.